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Land-atmosphere coupling: measurement, modelling and analysis

Ecohydrologic Processes Modify Urban Rainfall Intensification in Land-Atmosphere Simulations

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Key Points:

- We integrated sub-grid urban ecohydrologic processes into land-atmosphere simulations
- Ecohydrologic processes increase evapotranspiration, causing cooler and wetter surface conditions that feed into atmospheric processes
- Simulations that consider ecohydrologic processes produce more infiltration and rainfall patterns that more closely represent observations

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Ecohydrologic processes influence urban water and energy cycles including convective rainfall. Coupled land-atmosphere models, however, lack fine-scale representation of urban ecohydrology processes. This study integrates these processes into ensemble land-atmosphere simulations to assess changes in convective rainfall in Milwaukee, Wisconsin. We compare simulations using existing urban land surface representations to ones that resolve processes like tree canopy overhanging pavement and runoff routing from impervious to pervious surfaces. We find that ecohydrologic processes produce cooler and wetter conditions ahead of rainfall due to higher evapotranspiration, especially under weak atmospheric forcing. Simulated rainfall aligns more closely with observed rainfall when ecohydrologic processes are considered, especially in an area of known urban rainfall intensification; more of this rainfall infiltrates. Our findings highlight the importance of model representation of urban ecohydrology to better understand rainfall holistically and the importance of representing these processes for precipitation forecasts and for exploring future city-scale climate adaptation strategies.

Plain Language Summary Water movement in cities is complicated, and can affect surface fluxes and humidity in ways that are important for rainfall. Unfortunately, the computer models used to predict weather and climate do not account for this complicated movement. In this study, we look at ensembles of 10 rainstorms in Milwaukee, Wisconsin and investigated how water movement via tree canopy exchanges and runoff from paved surfaces to green spaces can influence city scale meteorology. Specifically, we examine how proper representation of these city exclusive processes feedback to change the rainfall simulated by these computer models. Simulations that include water movement caused cooler and more humid conditions in the city prior to rainfall especially during lake-breeze conditions. More water is also stored in the soil after storms. Those simulations also more closely matched actual rainfall measurements, especially downtown, where prior research has shown a rainfall "hotspot." Our findings suggest that small scale water movement is altering convective rainfall in our study region, and our results necessitate further analysis in cities across the world to discern if similar patterns emerge. In addition, the improved modeling capabilities shown here can improve precipitation forecast and can give city planners a more complete picture of how nature-based solutions can lead to more resilient cities.

1. Introduction

Urban spaces have expanded dramatically over the past 50 years (Thornton et al., 2023) and continue to grow (United Nations, 2018). Converting natural vegetation into a patchwork of buildings, roadways, sidewalks, and parks changes radiative (e.g., albedo, emissivity), thermodynamic (e.g., heat capacity), and aerodynamic (e.g., surface roughness) properties (Oke et al., 2017). This alters land-atmosphere energy fluxes, decreasing latent heat/evapotranspiration (ET), increasing sensible heat and heat storage, as well as increasing planetary boundary layer (PBL) depths (Arnfield, 2003; Grimmond & Oke, 1999). These changes create urban heat islands (UHI; Kong et al., 2021; Li et al., 2019; Oke, 1982)

Rainfall is also modified downwind and over cities, termed urban rainfall intensification (URI), first observed in the Midwestern US (Huff & Changnon, 1973). Subsequent studies show enhanced cloudiness (Vo et al., 2023), rainfall amounts (J. Liu & Niyogi, 2019; Shepherd, 2005; Sui et al., 2024), and rain intensity (e.g., Li et al., 2020; Yan et al., 2024) due to urbanization, though outcomes also depend on large scale atmospheric conditions (Liu et al., 2026). Causes of URI include: (a) UHI enhancement of vertical motion due to deeper and more well mixed

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PBLs, (b) increased surface roughness enhancing low-level water vapor convergence, and (c) anthropogenic aerosols enhancing cloud formation (Qian et al., 2022). Recent studies indicate that city size, density, and building scale are important factors (Torelló-Sentelles et al., 2024; Yang et al., 2024). URI can be stronger in coastal cities where moist on-shore winds during the daytime (e.g., land-sea breezes) help initiate convection and interact with large-scale forcings to generate more intense rainfall due to extra moisture (Sui et al., 2024; Yang et al., 2014).

Replacement of pervious landcover with impervious surfaces increases runoff and flooding (Hollis, 1975; Leopold, 1968; Wright et al., 2012), even where stormwater mitigation measures are implemented (Tran et al., 2024). Cities have responded by integrating nature-based solutions that prioritize vegetation and routing of rainfall to pervious surfaces to reduce runoff (Green et al., 2021; NASEM, 2019). The resulting increases in urban vegetation and soil moisture may enhance (reduce) latent (sensible) heat, creating more humid conditions. As cities implement more nature-based solutions, understanding their atmospheric impacts—including effects on rainfall—is needed.

We use the Weather Research and Forecasting (WRF) model to investigate how urban ecohydrology representation affects rainfall and surface hydrology. While atmospheric models like WRF are widely-used for examining urban climate and feedbacks (see Qian et al., 2022), their relatively coarse spatial resolutions ($\geq$1 km^2) cannot resolve finer-scale urban landscape features (e.g., sidewalk, driveway, trees, etc.) directly. As a result, studies often treat urban areas as uniform paved surfaces without vegetation (Chen et al., 2011; Liu et al., 2006). More detailed built environment representations like a single layer canyon model (Kusaka et al., 2001; Kusaka & Kimura, 2004) or a multi-layer building model (Kondo et al., 2005) are available but often represent all urban vegetation as grassland and neglect processes that interact across permeable-impermeable interfaces. Here, we use a recent urban land surface model called Noah-MP for Heterogenous Urban Environments (HUE; Alexander et al., 2024) to represent subgrid urban ecohydrology and explore impacts on city-scale rainfall and runoff. Noah-MP HUE explicitly represents subgrid impacts of lateral water transfers from one landcover type to another, making it unique among land-atmosphere models focused on urban rainfall.

Our study focuses on Milwaukee, Wisconsin. Milwaukee has both well-documented URI effects (Yang et al., 2013, 2014) and an aggressive urban greening plan (MMSD, 2013). Specifically, we aim to answer:

1. How does model representation of urban ecohydrology change the boundary layer prior to rainfall?
2. How do urban ecohydrologic representations impact simulated rainfall?
3. What are the city-scale effects of urban ecohydrologic processes on the water cycle?

2. Data and Methods

2.1. Model Configuration and Experimental Design

We used WRF v4.4.1 (Skamarock et al., 2019) with two one-way nested domains (9 km outer; 3 km inner) centered on the Milwaukee Metropolitan Sewerage District (MMSD) service area (Figure S1 in Supporting Information S1), using 51 vertical layers. We simulated 10 summertime rainstorms using hourly boundary conditions and Lake Michigan surface temperatures from ERA5 (Hersbach et al., 2020). Each simulation started >48 hr before rainfall onset, and the first 24 hr were discarded as spinup. Land surface states were initialized from long term offline simulations forced by downscaled ERA5 and Analysis of Record for Calibration (AORC) data (Abbasian et al., 2025; Fall et al., 2023).

Each storm was simulated using three urban scenarios within Noah-MP (He et al., 2023; Niu et al., 2011) throughout the MMSD area. Other WRF parameterizations remain unchanged (Tables S1 and S2 in Supporting Information S1). These scenarios are:

1. *Typical (default in Noah-MP)*: "Slab" parameterization with uniform surface roughness, permeability, and thermal parameters in urban grid cells and a single landcover elsewhere (Chen et al., 2011; Liu et al., 2006). National Land Cover Data set (NLCD; Dewitz, 2021) is used to define landcovers.
2. *Mosaic*: Each inner domain grid cell includes up to eight landcovers. Within each grid cell, each landcover is simulated independently and then averaged together based on fractional area using a "mosaic" scheme (Alexander et al., 2024; Li et al., 2013).
3. *HUE*: HUE represents tree canopy shading pavement and intercepting rainfall, and routes runoff from impervious surfaces (i.e., roofs) to surrounding pervious surfaces (i.e., yards). Runoff transfers and ET of

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overhanging tree cover are scaled by ratio landcovers: for example, transferring 2 mm of water from an impervious (25% of the grid cell) to turfgrass (16% of the grid cell), yields an additional flux of $2 \cdot \frac{.25}{.16} = 3.125$ mm to the turfgrass.

Differences between Mosaic and HUE are driven only by the HUE landcovers (ecohydrologic representations) and only represented in the MMSD service area (Alexander et al., 2025).

To assess model internal variability, we added 3 additional simulations per landcover scenario. These were generated by initializing 1–3 hr earlier than the original simulations with identical physics parametrizations. This creates a total of 120 simulations analyzed in this study.

2.2. Hydrometeorological Analysis

We analyzed simulated surface sensible and latent heat fluxes, potential temperature, specific humidity, and wind speed and direction throughout the PBL during afternoons before rain events (12–6 pm LST). Our analysis is centered over Milwaukee Mitchell International Airport to leverage observed PBL profiles of potential temperature, specific humidity, and winds (AMDAR; Moninger et al., 2003) extracted from aircraft takeoff and landing following Zhang et al. (2019). We grouped "Lake-Breeze" (i.e., winds from Lake Michigan toward Milwaukee) and "Non Lake-Breeze" pre-storm conditions using criteria from Laird et al. (2001) and created average PBL profiles.

Local-scale drivers from Lake Michigan can increase rainfall intensity in Milwaukee (Yang et al., 2014). Specifically, easterly low-level water vapor transport from Lake Michigan toward the city during storms, termed "Landward", can enhance rainfall compared to prevailing westerly transport, termed "Lakeward." Note that these are distinct processes: before-storm "Lake-Breeze" conditions can co-occur with during-storm "Lakeward" conditions and vice versa. We identified storms with predominantly "Landward" or "Lakeward" features using simulated near-surface (<500 m above ground level [agl]) low-level integrated water vapor transport ($\mathrm{IVT_{LL}}$; kg $\mathrm{m^{-1}\ s^{-1}}$) and low-level moisture flux convergence ($\mathrm{CON_{LL}}$; mm $\mathrm{hr^{-1}}$) during the hour before peak rainfall:

$$\mathrm{IVT_{LL}} = \int_{\mathrm{Surface}}^{500\ \mathrm{m}} U(z) \cdot q(z) \cdot \rho(z)\, dz \tag{1}$$

$$\mathrm{CON_{LL}} = -\left(\frac{\partial \mathrm{IVT}_x}{\partial x} + \frac{\partial \mathrm{IVT}_y}{\partial y}\right) \tag{2}$$

where $U(z)$ is the horizontal wind velocity at elevation z (m $\mathrm{s^{-1}}$), $q(z)$ is specific humidity ($\mathrm{kg_{water}\ kg_{air}^{-1}}$), and $\rho(z)$ is the density of air (kg $\mathrm{m^{-3}}$; Su & Smith, 2021). Positive values of both quantities indicate increasing moisture within a control volume. We categorized storms by the predominant flow direction before peak rainfall: "Landward" storms move water vapor in an easterly direction across most ensemble members, while "Lakeward" storms move water vapor in a westerly direction.

Simulated rainfall was compared to 19 MMSD rain gages and hourly gridded AORC rainfall, using mean error (ME):

$$\mathrm{ME} = \mathrm{mean}\left(M_t^i - O_t^i\right) \tag{3}$$

where O_t^i is the observed rainfall at a rain gage or pixel (mm), M_t^i is the simulated rainfall at the coincident pixel (mm), i denotes location, and t denotes time.

3. Results

3.1. Urban Land-Atmosphere Exchanges

Atmospheric profiles during "Non-Lake Breeze" events do not vary widely when comparing the Mosaic and HUE simulations to the Typical averages (Figures 1a–1d). All models present a cool bias compared to AMDAR observations (mean: 33.7°C). The largest differences for both temperature and humidity between simulations occur in the nearest 350 m, with warmer and less moist conditions occurring in Typical, while the Mosaic and HUE

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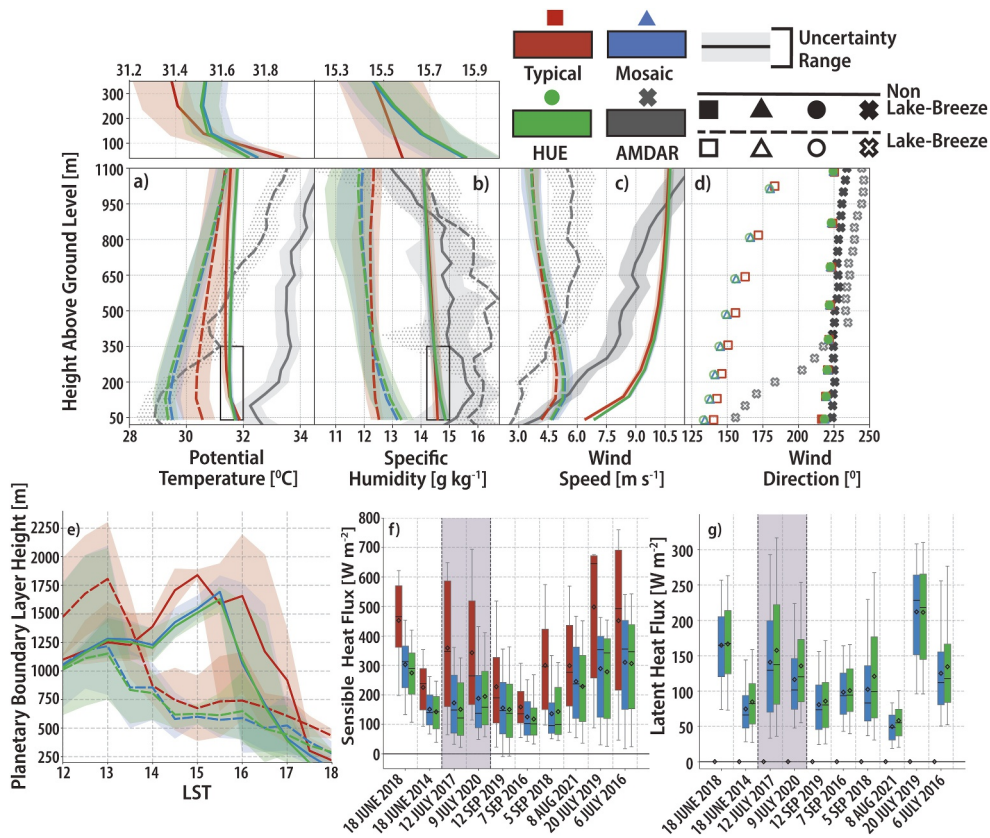


Figure 1. Average vertical profiles near Milwaukee Mitchell International Airport on the afternoon (12–6 p.m. LST) prior to rainfall of potential temperature (a), specific humidity (b), wind speed (c), and wind direction (d) disaggregated by afternoons with lake-breezes (12 July 2017 and 09 July 2020) and non lake-breeze events in Milwaukee, WI. Lines show the mean atmospheric profile averaged across all four ensemble members in the airport region, while the shading represents the range spanned by the individual ensemble-member averages. Boxes in panel (a) and (b) indicate zoom in region for the non lake-breeze events in the lowest 350 m. Black lines with shading/stippling indicate AMDAR flight averages and 95 percent confidence intervals, respectively. Average temporal evolution of planetary boundary layer heights (e) for both lake-breeze and non lake-breeze events with shading indicating range of average conditions. Afternoon sensible (f) and latent (g) heat fluxes from pixels near Milwaukee airport. Diamonds in box and whisker plots demarcate the average heat flux, whiskers contain the 5th and 95th percentiles, and gray shading indicate two lake-breeze events.

generate nearly identical cooler and moisture profiles. Simulated moisture profiles all lie within the uncertainty range of AMDAR throughout the lowest 1100 m, though are less variable than these observations. All simulations maintain a well-mixed PBL structure with realistic wind directions and profiles (Figures 1c and 1d). Simulated peak PBL heights in Typical (mean: 1,840 m) are higher on average compared to both Mosaic (mean: 1,690 m) and HUE (mean: 1,627 m; Figure 1e).

During "Lake Breeze" events, the role of urban heterogeneity and ecohydrology are more apparent due to the lack of larger scale forcing. AMDAR averages show a shallow intrusion of cool air from the surface to ~350 m agl (mean wind speed: 2.8 m s^{-1}) with complex wind shear in the lowest 800 m agl, while all simulations overmix the PBL with a deeper intrusion from Lake Michigan below 1100 m agl. Despite this overmixing, simulations depict a low jet structure with higher windspeeds in the lowest 350 m agl compared to aloft (Typical mean: 3.7 m s^{-1}, Mosaic mean: 4.0 m s^{-1}, HUE mean: 3.9 m s^{-1}; Figures 1c and 1d). Temperatures are well mixed in the lowest 200 m agl (mean: 29.0°C), while the Typical simulations are consistently warmer (mean: 30.5°C) below 200 m agl than Mosaic (mean: 29.4°C) and HUE (mean: 29.3°C; Figure 1a). Despite the lake breeze, these simulations are drier (means of 12.4, 12.9 and 13.1 g kg^{-1} for Typical, Mosaic, and HUE, respectively) than their non lake breeze counterparts, while AMDAR shows greater humidity (mean: 16.1 g kg^{-1}) in the lowest 200 m agl during lake breezes. PBL heights reveal much shallower depths during early afternoon, with a reduction around 13 LST

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due to onset of lake breezes, though Typical (mean: 910 m) over deepens the PBL compared to either Mosaic or HUE (mean of 680 and 669 m, respectively; Figure 1e).

Differences in atmospheric profiles for both "Lake-Breeze" and "Non Lake-Breeze" events are influenced by the partitioning of sensible and latent heat fluxes at the surface (Figures 1f and 1g). HUE enhances latent heat by 11 W m^{-2} on average compared to Mosaic, though this enhancement varies from −1 W m^{-2} to 26 W m^{-2} depending on event. HUE reduces sensible heat by an average of 134 W m^{-2} compared to Typical and 8 W m^{-2} compared to Mosaic with differences ranging from −8 W m^{-2} to 29 W m^{-2} between HUE and Mosaic, likely driven by available soil water.

3.2. Near Surface Hydrometeorology and Rainfall

Simulated rainfall totals at MMSD rain gage locations (Figure 2a) bracket raingage and AORC rainfall estimates in most cases, with HUE simulations producing closer means in half of our simulated cases. Milwaukee has a relatively unique URI signature, with a "bullseye" of enhanced rainfall just inland of Lake Michigan over downtown (Yang et al., 2013) instead of rainfall enhancement downwind of urban cores (Qian et al., 2022). When comparing the best performing ensembles within the URI region during "Landward" events, Typical simulations generally underestimate rainfall by 6 mm, while HUE better represents the URI bullseye on average (HUE: 2 mm overprediction) compared to Mosaic (4 mm overprediction; Figures 2b–2e). In "Lakeward" events, the URI bullseye is still present despite more uniform rainfall in observations (Figure 2f); Typical drastically underestimates rainfall (6 mm), while HUE and Mosaic simulations more closely follows the observed spatial pattern of rainfall, though there is still an under estimate (HUE: 2 mm, Mosaic: 1 mm). The most intense rainfall in simulations occurs over Lake Michigan (Figures 3v–3x), resembling a more traditional URI response.

"Landward" storms produce easterly IVT_{LL} from Lake Michigan (Figures 3a–3c) and a general increase in CON_{LL} west of Milwaukee (Figures 3g–3i). While IVT_{LL} is higher in Mosaic and Typical compared to HUE over the MMSD region (HUE: 30 kg m^{-1} s^{-1}, Mosaic: 38 kg m^{-1} s^{-1}, Typical : 36 kg m^{-1} s^{-1}), vertical cross sections reveal more moisture transport and more vertical motion on average over Milwaukee in HUE than others (Figures 3m–3o). Rainfall accumulations 1 hour after the initiation of this advection and convergence pattern (Figures 3s–3u) are similar in peak intensity between simulations (HUE: 15 mm, Mosaic: 16 mm, Typical: 16 mm), though the HUE rainfall is more widespread. "Lakeward" storms have higher IVT_{LL} south of Milwaukee, transported via southwesterly winds. The IVT_{LL} in Typical (86 kg m^{-1} s^{-1}) is higher than in HUE and Mosaic (63 kg m^{-1} s^{-1}; Figures 3j–3l). Weak CON_{LL} is present over much of the MMSD region (all simulations mean: 5 mm hr^{-1}; Figures 3j–3l), and vertical cross sections reveal more intense lift in Mosaic and especially HUE compared to Typical upwind and over Milwaukee. As a result, HUE and Mosaic still simulate higher rainfall (6 mm) compared to Typical (4 mm), despite lower IVT_{LL} and CON_{LL} (Figures 3p–3r). IVT_{LL}, transport direction, and CON_{LL} averaged over ensembles and the MMSD region for each storm are provided in Tables S5–S7 in Supporting Information S1.

3.3. Implications for Surface Hydrology

HUE simulations infiltrate more water (3 mm more on average) than Mosaic (Figures 4a and 4b) due to its ability to laterally transfer rain from impervious surfaces to pervious surface. Typical simulations do not infiltrate water in any urban pixel. Some variability in soil water change (Figure 4c) occurs due to spatial differences in rainfall. Runoff ratios are highest for Typical simulations: 1.0 in heavily urbanized regions (Figure 4d), approaching 1.0 in suburban regions, and 0.32 on average in peri-urban areas. In Mosaic simulations, ratios are much lower in heavily urbanized areas—below 0.8 more than 75% of the time. HUE runoff in the heavily urbanized (mean 0.66) and suburban (mean 0.48) are lower still.

4. Discussion

Prior studies have shown the importance of urban process representation in simulating temperature (Kusaka & Kimura, 2004; Salamanca et al., 2018; Sharma et al., 2016), wind (Chen et al., 2022; Droste et al., 2018; Flagg & Taylor, 2011), humidity (Huang & Song, 2023; Zixuan; Wang et al., 2021), and rainfall (Patel et al., 2023; Yang et al., 2024), while others have added multiple landcover types through mosaicking schemes in urban regions (Li et al., 2013; Sharma et al., 2016). Our HUE results indicate that subgrid representation of urban hydrologic and

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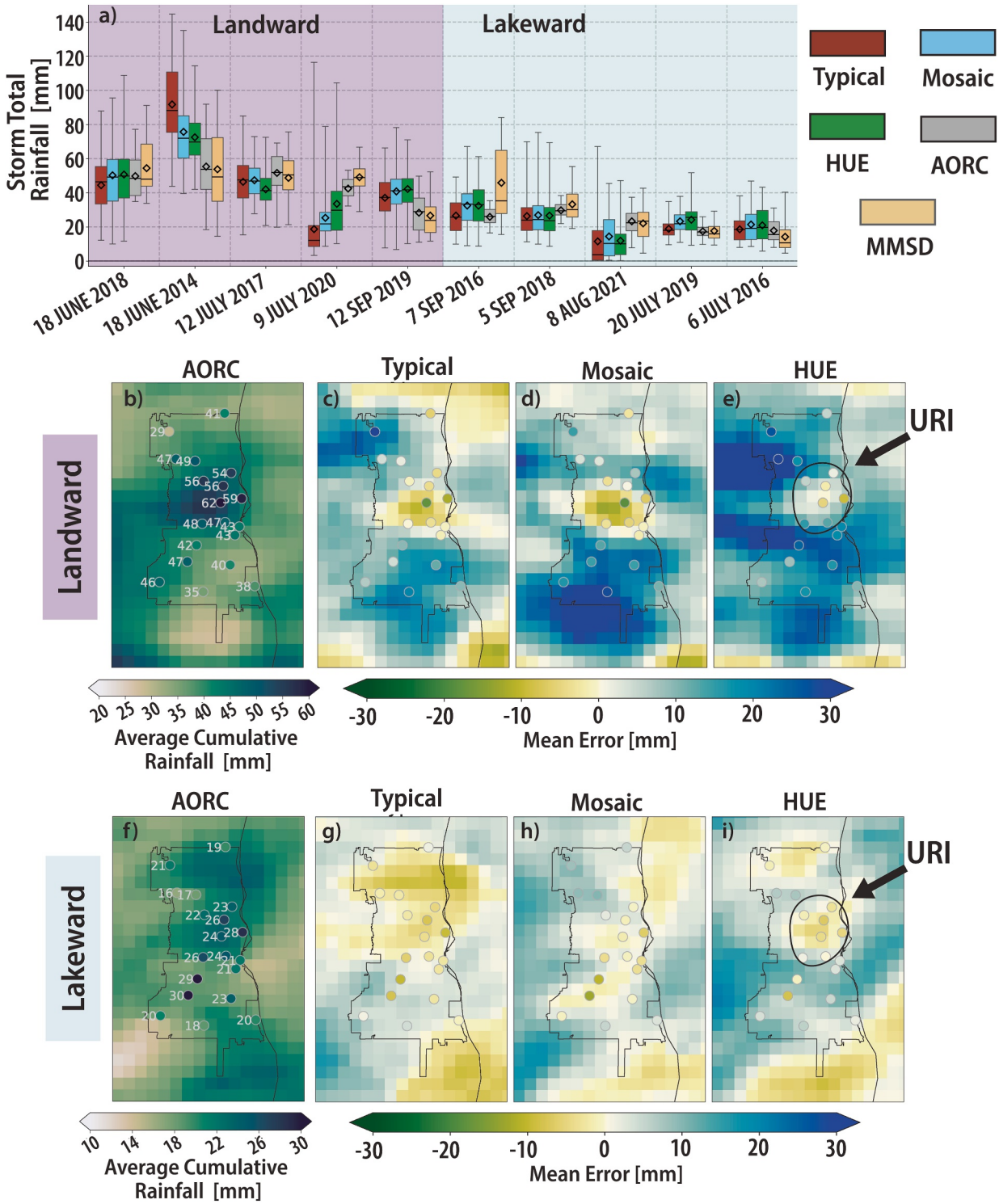


Figure 2. Box and whisker plots of storm total rainfall at milwaukee metropolitan sewerage district (MMSD) observation locations for Typical (red), Mosaic (blue), HUE (green), AORC (gray), and MMSD gages (gold) for all storms across all ensembles (a). Purple shading denotes "Landward" influenced and blue denotes "Lakeward" influenced events. Average cumulative rainfall totals for MMSD rain gages (circles) and AORC gridded precipitation for "Landward" (b) and "Lakeward" (f) storms. Mean error for "Landward" and "Lakeward" for lowest error ensemble members for each event in urban rainfall intensification (URI) region for Typical (c) and (g), Mosaic (d) and (h), and HUE (e) and (i). Demarcation of observed URI region is highlighted on (e) and (i).

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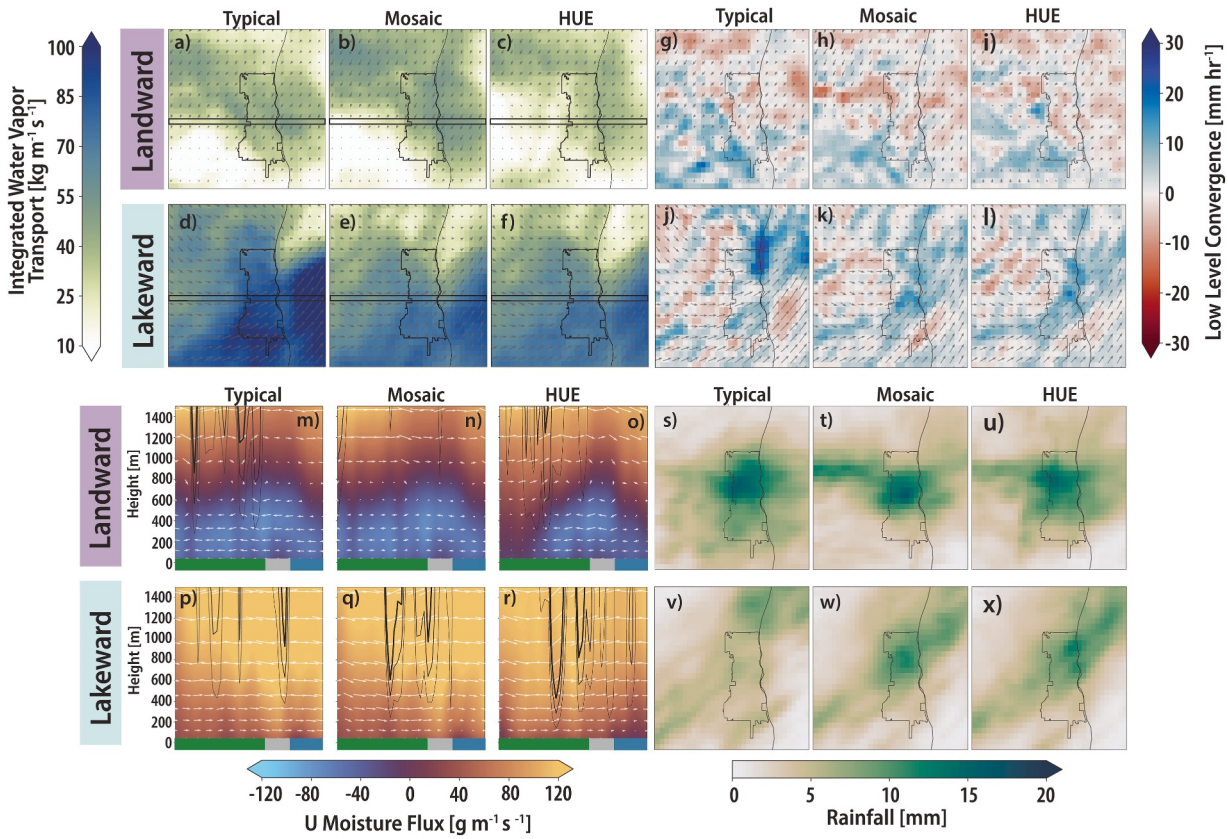


Figure 3. Low level (surface—500 m agl) IVT 1 hour before maximum 30 min rainfall intensity over the milwaukee metropolitan sewerage district region, with moisture vectors (a–f). Low-level convergence for the same period for with moisture vectors (g–l). Vertical cross section of U and W moisture flux from surface to 1,500 m agl, with moisture vectors and contours of vertical motion (thin: 1 g m^{-1} s^{-1}, medium: 2 g m^{-1} s^{-1}, thick: 3 g m^{-1} s^{-1}) with colored blocks denoting land surface (blue: Lake Michigan, gray: Milwaukee, green: non-urban) (m–r). Resulting maximum 30 min rainfall (s–x). First and third rows are averages of ensembles with "Landward" events, while second and fourth rows are "Lakeward" events.

vegetation interactions among landcover types are also important for real and simulated rainfall processes due to complex land-atmosphere interactions, especially during weak atmospheric forcing conditions.

Urban vegetation is a potentially important URI driver in Milwaukee through two intermingled processes: (a) increased latent heat and humidity and (b) reduced sensible heat and temperature; both occur throughout the lower atmosphere prior to rainfall, with larger impact when large-scale atmospheric forcing is weak. Better representation of tree cover in HUE, which includes urban tree canopy overhanging pavement (see details in Alexander et al., 2024) results in higher ambient atmospheric humidity and cooler temperatures. This additional process is especially active in the highly urbanized downtown that is subject to URI (Yang et al., 2013). As a result, HUE during "Lake-Breeze" events more closely match AMDAR atmospheric profiles of temperature and humidity than our other simulations, though errors remain. These errors could be related to PBL exchange issues (Avolio & Federico, 2018; Cohen et al., 2015; Hernández et al., 2024), which are particularly complex in heterogeneous urban regions (Baklanov et al., 2011; Barlow, 2014).

Subtle changes in humidity and buoyancy during rainfall causes more IVT_{LL} during "Landward" storms, and together with moisture advection over the city itself results in more realistic rainfall peak intensities and storm totals, especially in the URI region. The increased moisture advection aloft and vertical motion over Milwaukee from HUE indicates that seemingly small energetic and near surface meteorologic conditions caused by eco-hydrologic processes can play important roles in rainfall processes, especially in locally-forced storms. While the proximity to Lake Michigan makes Milwaukee relatively unique, the increased low-level moisture from urban vegetation, paired with the urban changes in roughness and heat, presents a mechanistic pathway where-in mesoscale convective rainfall is enhanced by inflow and lift, therefore potentially increasing extreme rainfall. Further testing in multiple locations under a variety of rainfall forcings is needed to determine the uniqueness of

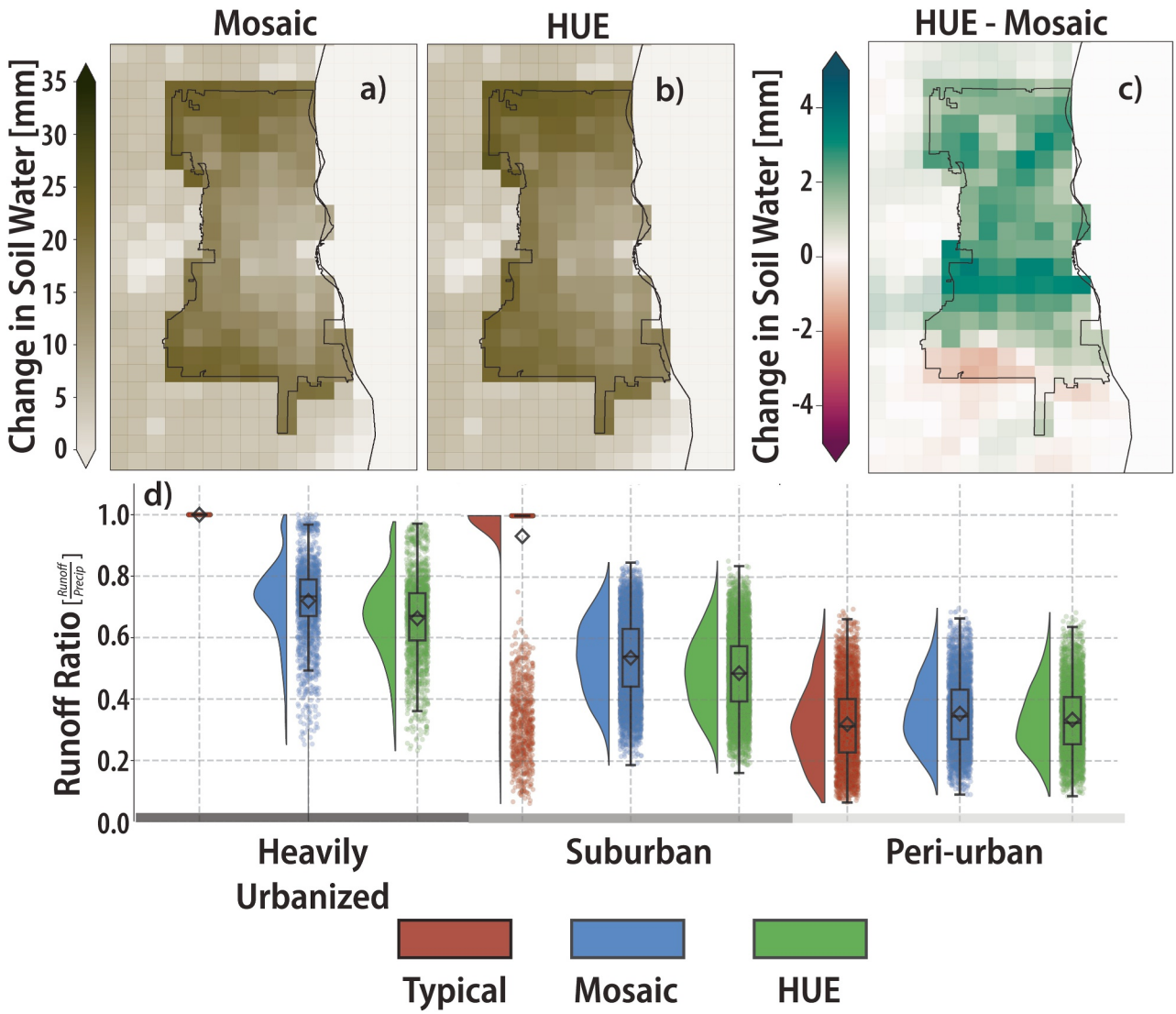


Figure 4. Ensemble average change in soil water during rainstorms for Mosaic (a) and HUE (b) and difference between HUE and Mosaic (c). Box and whisker plots of runoff ratio—the amount of runoff over a 30-min period divided by the amount of precipitation over the same period—when at least 5 mm of rainfall fell during a 30-min period, disaggregated by NLCD landcover type (d) for all ensembles. Note that the suburban group contain a majority of impervious only landcovers, with a few urban green space landcovers that may infiltrate water. Diamonds in box and whisker plots demarcate the mean.

this mechanism (e.g., whether it is more common in coastal cities than their inland counterparts), and will require examination of multi-event ensembles like those used here, especially since similar changes at the land surface can lead to a wide range of outcomes (Tuttle & Salvucci, 2016).

Many studies of urban climate (see Qian et al., 2022 for thorough review) and urban hydrology (Fletcher et al., 2013; Salvadore et al., 2015) ignore the feedbacks between atmospheric and ecohydrologic processes. While the Mosaic scheme in our work gives results similar to HUE for atmospheric variables under strong atmospheric conditions (Figures 1a–1d), differences in locally forced ambient conditions and rainfall (Figure 2b–2e), surface fluxes (Figures 1f and 1g), and especially soil moisture and infiltration (Figure 4) are large. The increased infiltration is critical, as trees in urban areas generally use more water due to higher vapor pressure deficits compared to natural ecosystems (Clark & Kjelgren, 1990; Zipper et al., 2017). Urban trees under water stress potentially eliminate one of the intended benefits of additional greenery (e.g., evaporative cooling) as water stressed plants can be significantly warmer (Gräf et al., 2021), and may lead to warming at city scales (Alexander et al., 2025; Schlaerth et al., 2023).

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The ecohydrologic processes in HUE allow for first-of-its-kind testing of climate adaptation initiatives at the city scale, simulating hydrologic, vegetative and atmospheric interactions together under current and future climate scenarios. Examples of these initiatives range from planting urban vegetation to more complex hydrologic interventions (Choi et al., 2021) that infiltrate rainfall and reduce/slow runoff (Green et al., 2021). Our findings suggest that (a) urban evapotranspiration will increase under this plan, and (b) greater infiltration will result in less soil water stress-induced curtailment of evapotranspiration. This facilitates anticipated co-benefits of greening plans, including temperature reduction (Bowler et al., 2010; Knight et al., 2021) and improvements in human health (Chen et al., 2021; Kondo et al., 2018). At the same time, climate change is expected to increase dry stability (Frierson, 2006) and more extreme heavy rainfall due to a shift to warmer and moister conditions (Rasmussen et al., 2020; Wang & Moyer, 2023). As cities adapt to heat and enhanced flood risks via widespread implementation of so-called nature-based solutions, there could be unexpected instances of more extreme rainfall in and downwind of cities despite the aforementioned benefits of cooling. Thus, it is imperative that models like HUE are used to evaluate climate adaption initiatives holistically to understand climate changes and potential tradeoffs.

5. Conclusions

We compared a slab urban model (Typical), a mosaic urban approach (Mosaic), and simulations that included urban hydrology and vegetation processes (HUE) to illustrate the importance of fine-scale urban ecohydrologic interactions on rainfall and runoff generation processes. Specifically, we represented the effects of trees overhanging pavement (shading, interception, and increased transpiration footprint) and routing of runoff from impervious surfaces to pervious surfaces. In our simulations, HUE generated cooler and wetter ambient conditions in the lower atmosphere, which led to better agreement with observed rainfall and aircraft-derived profiles during weak atmospheric forcing. We found that HUE changes low-level water vapor transport prior to maximum rainfall rates and improves representation of URI. Meanwhile, rainfall in HUE simulations is more likely to infiltrate than in than Mosaic and Typical simulations, especially in heavily urbanized locations, due to sub-grid routing of rainfall from impervious to pervious landcovers (i.e., rooftops to yards). Our findings suggest that enhanced infiltration and the propensity to convert rainfall into atmospheric water vapor via evapotranspiration can cool an urban area but opens new questions on how cities continue to modify rainfall.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The ERA5 global reanalysis data are available from Hersbach et al. (2020). The WRF model with Noah-MP HUE integrated, observations from MMSD rain gages are provided in GAaronAlexander (2026). AMDAR data used for this study is hosted by MADIS and accessed through this service portal (https://madis-data.ncep.noaa.gov/madisPublic1/data/archive/). AORC data was obtained from Abbasian et al. (2025) as part of their FUSION data set.

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